

ConSynth-X: Benchmarking Construction CV Under Extreme Field Conditions

Project Progress Report

Viet Huy Duong
vduong1@kent.edu

Kent State University

April 8, 2026

Outline

- 1 Problem & Motivation
- 2 Datasets
- 3 Augmentation Pipelines
- 4 Benchmarking Framework
- 5 VQA Safety Results
- 6 Description Task Results
- 7 Augmentation Quality Evaluation
- 8 Project Architecture
- 9 Downstream Detection – Sensitivity Analysis
- 10 Current Status & Next Steps

Problem Statement

Goal: Evaluate how robust construction computer vision models are under extreme field conditions.

Why this matters:

- Construction sites face rain, snow, fog, low light, glare
- Models trained on clean data fail in the field
- No standardized benchmark for these conditions

ConSynth-X provides:

- 1 Synthetic augmentation pipelines (5 methods)
- 2 Multi-label taxonomy of 40+ conditions
- 3 VLM benchmarking framework (11 models)
- 4 Image quality evaluation (FID, DINO structure)

6 Condition Categories

Weather (rain, snow, fog)

Lighting (night, glare)

Scale (small/distant)

Image Quality (blur, noise)

Scene Complexity

Domain Shift

Base Datasets

ConstructionSite10k

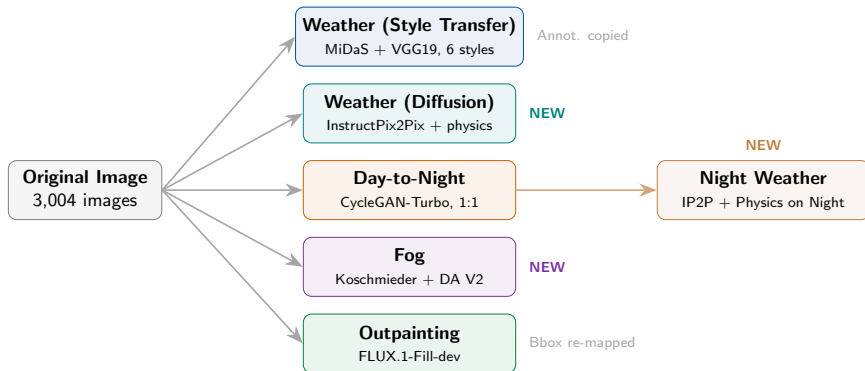
- Format: HuggingFace Arrow
- 3 classes: excavator, rebar, worker_with_white_hard_hat
- Test set: **3,004** images
- Rich annotations: bounding boxes + 4 safety rules + captions

SODA Dataset

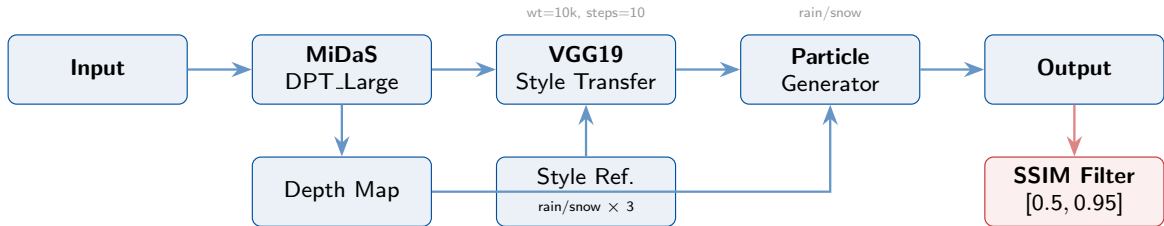
- Format: Pascal VOC (XML)
- **15** object classes
- Total: **19,846** images
- Standard VOC train/val/test splits

Condition	Method	Constr. Site	SODA
Original	Baseline	3,004	19,846
Weather (style)	Neural Style Transfer	10,266	47,169
Weather (diff.)	InstructPix2Pix	10,382	–
Night	img2img-turbo	3,004	19,846
Night Weather	Night → IP2P + Physics	Pending	–
Fog	Koschmieder + DA V2	3,003	–
Small	FLUX Outpainting	1,323	1,001
Total		30,982+	87,862

Overview of 5 Augmentation Pipelines



Weather Augmentation Pipeline



Construction Site (3,004 original)

SODA (19,846 original)

Style	After Filter	Style	After Filter
rain_0 / rain_1 / rain_2	2,211 / 1,243 / 1,208	rain_0 / rain_1 / rain_2	8,392 / 8,059 / 9,541
snow_0 / snow_1 / snow_2	2,159 / 1,649 / 1,796	snow_0 / snow_1 / snow_2	9,818 / 7,485 / 3,874
Total	10,266	Total	47,169

NEW: Diffusion-Based Rain/Snow (InstructPix2Pix)

Model: InstructPix2Pix
(timbrooks/instruct-pix2pix)

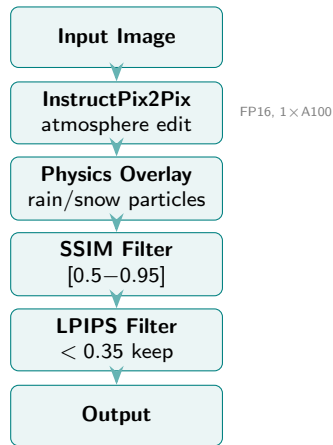
Why IP2P? Replaces neural style transfer with direct text-guided image editing while preserving scene structure.

Hybrid approach: IP2P atmosphere + physics overlay

- IP2P: dark sky, wet ground, frost, snow cover
- Physics: rain streaks (3 layers, 2800–5200) + fog overlay
- Snowflakes: 3 layers, 2100–3300, screen blend

Key parameters:

	Rain	Snow
image_guidance	1.5	1.5
guidance_scale	10.0	8.0
steps	30	30



Post-filter: SSIM rejects unchanged/corrupted; LPIPS rejects structural loss ≥ 0.35 .

Weather Augmentation: Visual Comparison

Original



Rain
SSIM=0.78, LPIPS=0.17

Style Transfer



IP2P Diffusion



Rain (subtle)
SSIM=0.85, LPIPS=0.10



Night Weather Augmentation (IP2P + Physics)

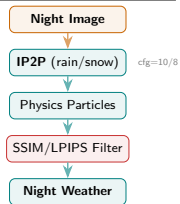
Pipeline: Night $\rightarrow$ IP2P $\rightarrow$ Physics

Particles

Compound conditions (rain+night, snow+night) are common on real sites but absent from training data.

- 1 Start from **CycleGAN night** image
- 2 IP2P edits atmosphere (night prompts)
- 3 Physics overlay: rain streaks / snowflakes
- 4 SSIM/LPIPS filter (vs. *original*)

Param	Rain	Snow
guidance_scale	10.0	8.0
img_guidance	1.5	1.5
steps	30	30
SSIM range	[0.6, 0.95]	[0.5, 0.95]
LPIPS	<0.35	-



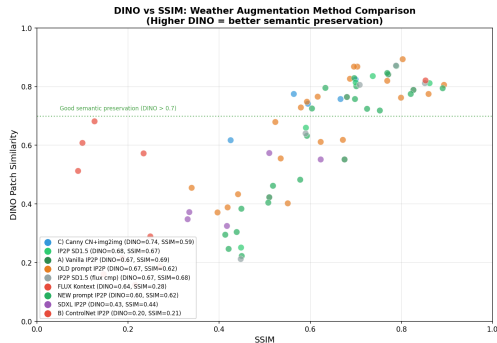
Why Not SSIM? DINO vs SSIM for Weather Augmentation

Problem: SSIM measures **pixel-level** similarity — it penalizes *desired* changes (darker sky, rain streaks, color shift) equally with *unwanted* changes (hallucinated objects, structural damage).

DINOv2 Patch Similarity measures **semantic structure** — objects, layout, spatial composition — without penalizing atmospheric changes.

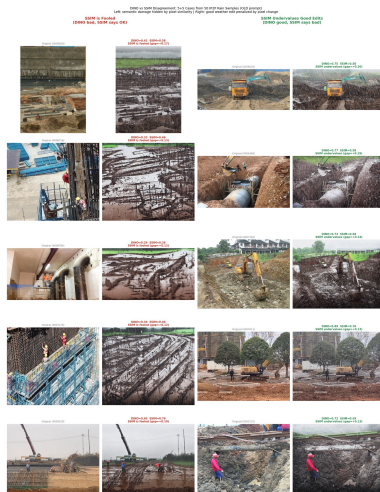
Method	DINO↑	SSIM↑
Canny CN+img2img	0.744	0.590
Vanilla IP2P	0.674	0.692
FLUX Kontext	0.639	0.281
SDXL IP2P	0.435	0.443
ControlNet IP2P	0.196	0.214

DINO ranks Canny CN+img2img #1 (strong weather + preserved objects), while SSIM ranks Vanilla IP2P #1 (subtle edit $\approx$ less change).



DINO–SSIM scatter across all methods. Higher DINO = better semantic preservation. FLUX has low SSIM but decent DINO (global=0.92).

DINO vs SSIM Disagreement: When SSIM Fails



Left (red): SSIM says OK but DINO says bad — ground texture changes hidden by similar brightness.

Right (green): SSIM says bad but DINO says good — strong weather edit preserves all objects.

NEW: Fog Augmentation (Koschmieder Model)

Model: Depth Anything V2 (Small) + Koschmieder atmospheric scattering

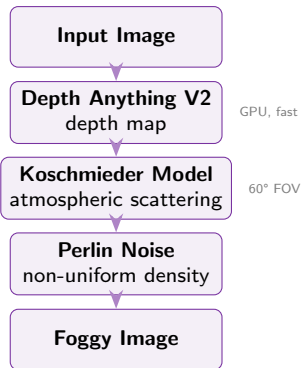
Approach: Physics-based fog synthesis using per-pixel depth-aware attenuation with Perlin noise for non-uniform density.

3 Intensity Zones:

Label	Visibility (m)	Description
heavy	300–500	Background obscured
medium	500–750	Objects recognizable
light	750–1000	Subtle morning haze

Key features:

- Smoothstep layer blending (no horizontal banding)
- Multi-octave Perlin noise for realism
- Fog color: 210–245 (avoids white washout)
- All annotations preserved



Output: $3 \times 1,001 = 3,003$ images

(heavy + medium + light, Construction Site)

Arrow format with fog_label, fog_visibility metadata columns.

Day-to-Night Augmentation

Model: CycleGAN-Turbo (day_to_night checkpoint)

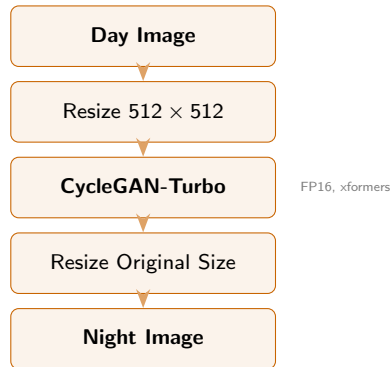
Pipeline:

- 1 Load day image $\rightarrow$ resize to 512×512
- 2 Forward pass through CycleGAN-Turbo
- 3 Resize output back to original resolution (LANCZOS)
- 4 Save night image + copy annotations directly

Key properties:

- **1:1 mapping** – every image gets one night variant
- FP16 inference with xformers attention
- Annotations **copied unchanged** (pixel-level transform)

Output: 3,004 (Construction Site) + 19,846 (SODA)



Outpainting – Scale/Distance Augmentation

Model: FLUX.1-Fill-dev (bfloat16)

Idea: Expand canvas around original image → objects appear smaller/more distant.

Scale factor: Truncated Gaussian

$$s \sim \mathcal{N}(\mu=0.25, \sigma=0.05), \quad s \in [0.2, 0.4]$$

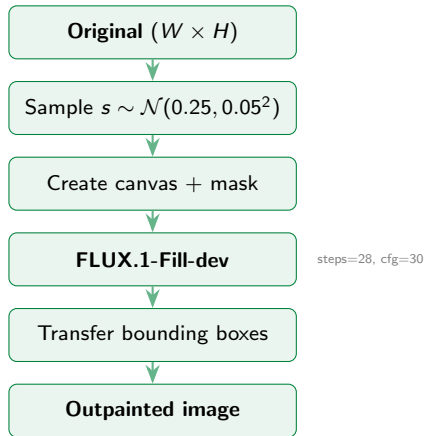
Canvas size: $(1 + 2s) \times$ original on each axis.

Bbox transfer:

$$x'_1 = \frac{x_1 \cdot W \cdot r + p_x}{W'}, \quad y'_1 = \frac{y_1 \cdot H \cdot r + p_y}{H'}$$

where r = resize factor, (p_x, p_y) = paste offset

Prompt: *“an outdoor construction site with buildings, roads, and open sky in the background”*



Dataset	Output
Construction Site	1,323
SODA	1,001

VLM Benchmarking – Models & Tasks

11 Vision-Language Models:

Model	Size	Status
InternVL2.5	26B	Done
Gemma-3	27B	Done
Qwen3-VL	8B	Done
Phi-4-multimodal	6B	Done
GPT-4o-mini	API	Done
LLaVA v1.5	7B	Failed
Qwen2.5-VL	7/32/72B	Pending
Llama-3.2-Vision	11B	Pending
GPT-4o	API	Pending

4 Benchmark Tasks:

1 Description (0-shot captioning)

Metrics: BERTScore, BLEU, ROUGE-L, METEOR, CLIPScore

2 VQA Safety (5-shot, 4 rules)

Metrics: Per-rule P/R/F1 + bbox IoU

3 VQA Rule 1 (binary Yes/No)

PPE compliance detection

4 Object Detection

Normalized bbox, grid-based IoU

Conditions: original × weather × night × small

Safety Rules for VQA

Rule 1 – PPE Compliance

- Hard hats, proper clothing, safety shoes
- High-vis vests at night
- Face shields when cutting/welding

Rule 2 – Safety Harness

- Required at height $\geq 3\text{m}$
- Without edge protection

Rule 3 – Edge Protection

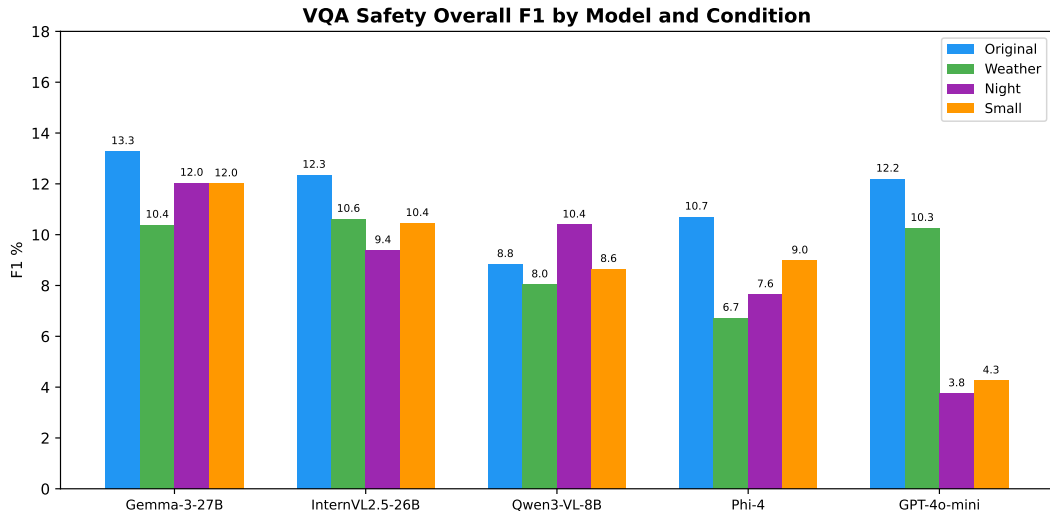
- Guardrails, fences required
- Underground projects $\geq 3\text{m}$ depth

Rule 4 – Excavator Safety

- Workers in blind spots
- Within operation radius

Output format: JSON with rule ID, reason, bounding box $[x_1, y_1, x_2, y_2]$ in normalized $[0, 1]$ space.

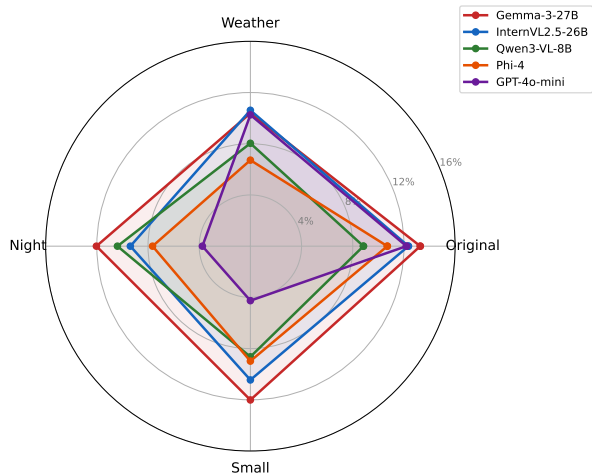
VQA Safety – Overall F1 by Model and Condition



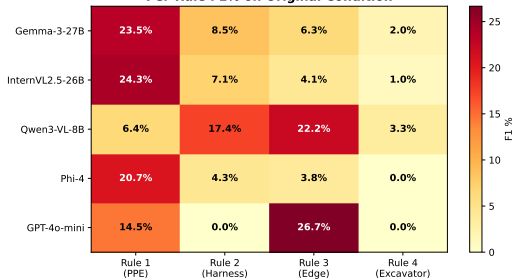
5-shot prompting (Phi-4: 2-shot due to memory). LLaVA-1.5-7B omitted (100% parse errors). Now includes **Gemma-3-27B**.

VQA Safety – F1 Radar Across Conditions

VQA Safety F1 by Condition



Per-Rule F1% on Original Condition



VQA Safety – Key Observations

Model Comparison (6 models evaluated):

- **Gemma-3-27B**: Best F1 (13.3%), most robust across conditions (F1 12.0% on night/small)
- **InternVL2.5-26B**: Very high recall (70–84%) but extremely low precision (5–7%) – over-predicts violations
- **Qwen3-VL-8B**: Best bbox IoU (28.9%) with balanced P/R – best localization
- **Phi-4** (2-shot): F1=10.7%, competitive with GPT-4o-mini; 5-shot OOMs on 1×A100
- **GPT-4o-mini**: ~50% error rate (model_error + missing responses)
- **LLaVA-1.5-7B**: 100% parse errors – cannot produce JSON

Condition Impact:

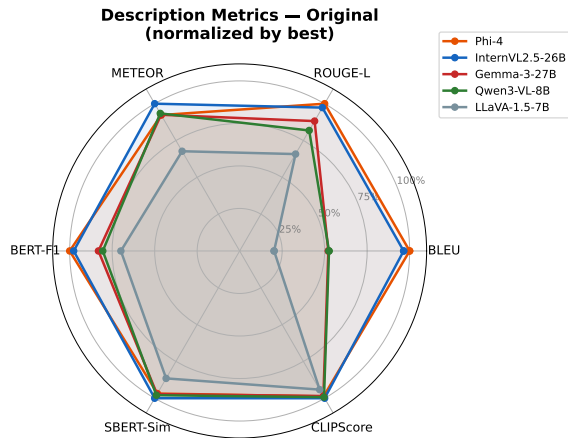
- **Night** degrades GPT-4o-mini most (F1: 12.2→3.8%)
- Gemma-3 and InternVL2.5 are **robust** across conditions
- All models: **very high false positive rate**

Per-Rule F1% (original, best model):

Rule	Best F1	Best Model
R1 (PPE)	24.3%	InternVL2.5
R2 (Harness)	17.4%	Qwen3-VL
R3 (Edge)	26.7%	GPT-4o-mini
R4 (Excavator)	12.9%	Qwen3-VL

Takeaway: Even the best model achieves ~0.30 IoU. Current VLMs lack domain knowledge for reliable safety inspection.

Description Task – Model Profiles (Original)



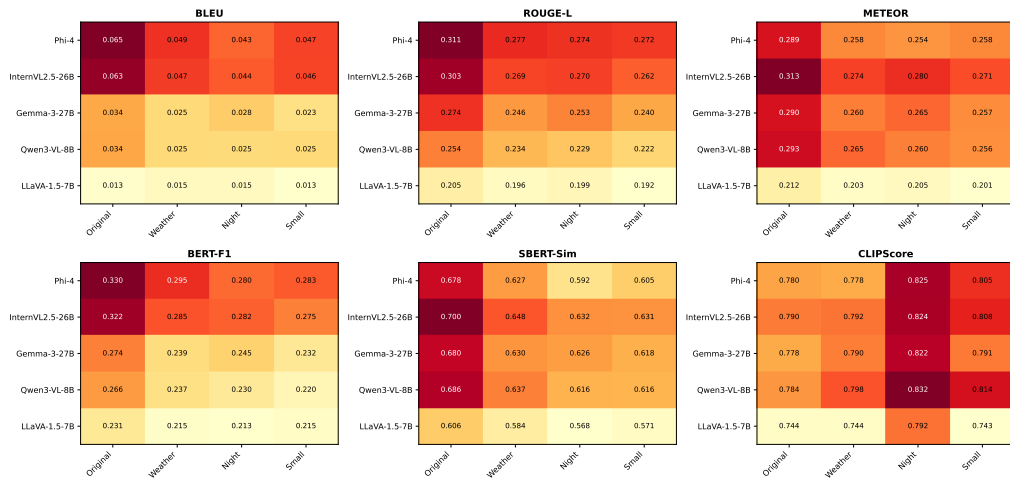
5 models evaluated (GPT-4o pending):

- **Phi-4** (6B) leads: BLEU 0.065, BERT-F1 0.330 – *competitive with InternVL 26B*
- **InternVL2.5-26B**: Best ROUGE-L (0.303)
- **Gemma-3-27B**: Between Qwen3 and InternVL
- **CLIPScore** high across all (0.74–0.83) – descriptions are semantically relevant
- **LLaVA-1.5-7B**: Weakest across all metrics

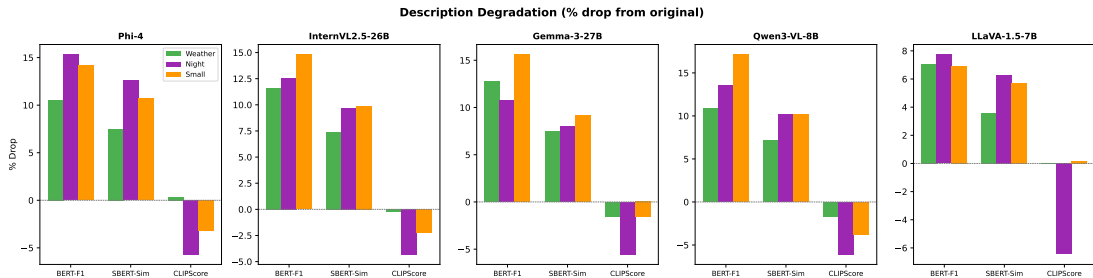
Note: CLIPScore consistently *increases* on night/small conditions (0.80–0.83 vs 0.78 original), likely because augmented images have simpler backgrounds.

Description Task – All Metrics Heatmap

Description Metrics — All Models x Conditions



Description Task – Degradation Under Augmentation



Positive = performance drop from original. SBERT-Sim and CLIPScore show minimal degradation, confirming augmented images remain semantically consistent.

Augmentation Quality – FID & DINO Structure Similarity

Fréchet Inception Distance (FID)

Measures distributional distance between original and augmented images in Inception-v3 feature space.

$$\text{FID} = \|\mu_r - \mu_g\|^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2})$$

- **Lower = closer** to real distribution
- Captures both quality and diversity
- Computed per augmentation condition

DINO Structure Similarity

Measures structural preservation using self-supervised DINOv2 features.

- Extract patch-level features from DINOv2-ViT
- Compute cosine similarity between original and augmented feature maps
- **Higher = better** structural preservation
- Captures spatial layout, object structure, and scene composition

Why DINO? Self-supervised features correlate with human perception of structural similarity better than pixel-level metrics.

Goal: Verify augmented images preserve realism (FID) and scene structure (DINO) while introducing meaningful visual variation.

Quality Evaluation – Per-Condition Analysis

FID Scores (lower = closer to original):

Condition	FID ↓	Images
rain_0	27.8	2,211
snow_0	39.6	2,159
rain_2	40.8	1,208
rain_1	43.6	1,243
snow_2	50.5	1,796
small	60.2	1,323
snow_1	69.9	1,649
night	97.8	3,004

DINO Structural Similarity (higher = better):

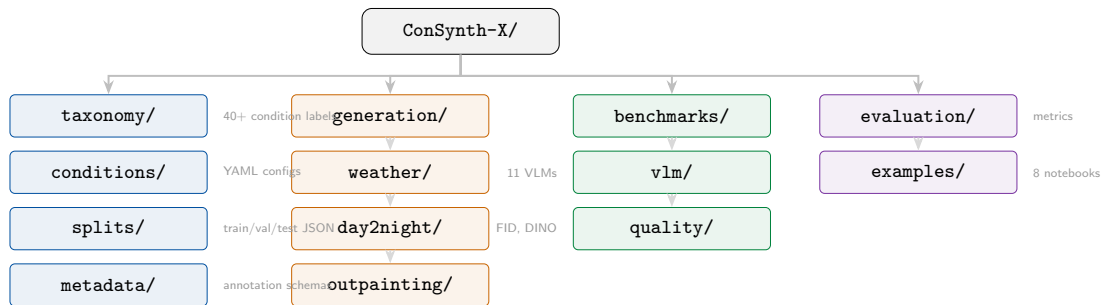
Condition	Mean ↑	Std
rain_0	0.893	0.103
rain_2	0.846	0.167
snow_0	0.842	0.138
rain_1	0.829	0.190
night	0.754	0.100
snow_2	0.753	0.144
snow_1	0.744	0.176
small	0.729	0.113

DINOv2-ViT-S/14, cosine similarity of patch features.

Key Findings:

- **Rain_0** best quality: lowest FID (27.8) and highest DINO (0.893) – subtle, realistic effect
- **Night** largest distribution shift (FID 97.8) but decent structure preservation (DINO 0.754)
- **Small** (outpainting) lowest structural similarity (0.729) – expected due to scene expansion
- Snow styles show more variation than rain (FID 39.6–69.9 vs 27.8–43.6)

Repository Structure



SSIM/LPIPS Threshold Sensitivity – Experimental Design

Goal: Determine optimal SSIM/LPIPS thresholds for quality filtering of synthetic augmentation data.

Pipeline:

- 1 Generate rain/snow via IP2P + physics overlay
- 2 Filter by $\text{SSIM} \in [\tau, 0.95]$ and $\text{LPIPS} < \lambda$
- 3 Combine filtered augmented + original data
- 4 Train YOLOv8n (20 epochs, seed=42)
- 5 Evaluate mAP@0.5 on held-out test set

Training setup:

- Model: YOLOv8n, imgsz=640, batch=16
- 3 classes: excavator, rebar, worker
- Val: 1,000 images from test split

7 threshold configurations:

Config	SSIM _r	SSIM _s	LPIPS
baseline	—	—	—
loose	0.40	0.40	0.50
moderate	0.50	0.50	0.40
current	0.60	0.50	0.35
strict	0.65	0.60	0.30
tight	0.70	0.65	0.25
no_lips	0.60	0.50	—

r = rain, s = snow. Baseline uses original data only (no augmentation).

Sensitivity Results – YOLOv8n Detection Performance

Config	Train	mAP@0.5	mAP@.5:.95	Prec.	Recall	Excavator
baseline	7,009	0.562	0.338	0.674	0.558	0.906
loose [†]	17,391	0.535	0.344	0.589	0.579	0.920
moderate [†]	17,391	0.535	0.344	0.589	0.579	0.920
current	17,390	0.547	0.339	0.617	0.558	0.916
strict	16,379	0.538	0.349	0.583	0.580	0.915
tight	14,948	0.542	0.344	0.613	0.576	0.909
no_lpip [†]	17,391	0.535	0.344	0.589	0.579	0.920

[†] **Identical results** – see next slide for explanation.

Per-class AP@0.5: Excavator consistently high (0.91+). Rebar struggles (0.19–0.26). Worker stable (0.45–0.52).

Sensitivity Analysis – Limitations Found

Problem: Pre-filtered Arrow data

The Arrow batch files used for training contain **only images that already passed** the current threshold ($\text{SSIM} \geq 0.60$ rain, ≥ 0.50 snow).

Consequence:

- Looser thresholds (loose, moderate) cannot recover dropped images
- `no_lpips` produces identical data to `loose` (LPIPS filter was already applied)
- Only `strict` and `tight` actually filter differently

Evidence:

Config	Rain	Snow
loose/mod/no_lpips	3,627	6,755
current	3,626	6,755
strict	3,048	6,322
tight	2,259	5,680

Additional issues:

- ❶ **Snow has no LPIPS scores** – LPIPS was never computed for snow diffusion, making `no_lpips` config meaningless (65% of augmented data)
- ❷ **Val set is clean images only** – baseline outperforms all augmented configs because the test set has no weather conditions
- ❸ **Only 20 epochs** – may not be sufficient for convergence with larger datasets (17k vs 7k)

Fix plan:

- Re-run filtering from raw meta CSVs + **unfiltered** Arrow batches
- Compute LPIPS for snow
- Add weather-augmented val set
- Increase to 50 epochs

Current Status & Next Steps

Completed:

- ✓ 5 augmentation pipelines total
- ✓ VLM benchmark framework (11 models)
- ✓ VQA Safety: 6 models $\times$ 4 conditions
- ✓ Description: 5 models $\times$ 4 conditions
- ✓ Augmentation quality (FID, DINO)
- ✓ Gemma-3-27B, Phi-4 integrated
- ✓ Fog pipeline (Koschmieder + DA V2)
- ✓ Diffusion rain/snow (IP2P, 10,382 images)
- ✓ SODA day-to-night (19,846 images)
- ✓ Sensitivity analysis v1 (7 configs, YOLOv8n)

In Progress / Next:

- **Fix sensitivity:** re-run from unfiltered Arrow data (current results invalid for loose/moderate)
- Compute LPIPS for snow diffusion
- Add weather-augmented val set
- Re-train with 50 epochs
- Run remaining VLMs (Qwen2.5, Llama-3.2, GPT-4o)
- FID & DINO eval on fog + diffusion weather

Planned:

- Per-condition robustness metrics (AP)
- Failure-case visualizations
- Paper with reporting tables

Thank You

Questions?

Repository: ConSynth-X/
Viet Huy Duong – vduong1@kent.edu
Kent State University